

Multiple Choice (10 marks)

1. Which of the following statements about a remote procedure call is true?
 - (a) It is used to call procedures with addresses that are farther than 2^{16} bytes away.
 - (b) It cannot return a value.
 - (c) It cannot pass parameters by reference.
 - (d) It cannot call procedures implemented in a different language.
2. Which is a valid expression in Python 3.5?
 - (a) `sort('ab')`
 - (b) `sorted('ab')`
 - (c) `"ab".sort()`
 - (d) `1/0`
3. Let `l = [1,2,2,3,4]`. In Python 3, what is a possible output of `set(l)`?
 - (a) `{1}`
 - (b) `{1,2,2,3,4}`
 - (c) `{1,2,3,4}`
 - (d) `{4,3,2,2,1}`
4. Which of the following is not a sentence that is generated by the grammar $A \rightarrow BC$, $B \rightarrow x|Bx$, $C \rightarrow B|D$, $D \rightarrow y|Ey$, $E \rightarrow z$?
 - (a) `xyz`
 - (b) `xy`
 - (c) `xxzy`
 - (d) `xxxxy`
5. Which of the following statements is FALSE about memory reclamation based on reference counting?
 - (a) Reference counting is well suited for reclaiming cyclic structures.
 - (b) Reference counting incurs additional space overhead for each memory cell.
 - (c) Reference counting is an alternative to mark-and-sweep garbage collection.
 - (d) Reference counting need not keep track of which cells point to other cells.

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `['Hi!'] * 4` produces the list `['Hi!', 'Hi!', 'Hi!', 'Hi!']`.
7. True or False: Operator precedence in a programming language can be effectively specified using a context-free grammar.
8. True or False: The number of distinct functions mapping a finite set with m elements into a finite set with n elements is n^m .
9. True or False: Church's thesis was first proven by Alan Turing, establishing the equivalence of computable functions.
10. True or False: Interrupts can indicate the completion of an I/O operation, signaling the processor to handle the next task.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the given number is a perfect square or not.
12. Write a function to calculate perimeter of a parallelogram.

End of Paper